Experiment 7 - Bragg's Law

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1 Background

In 1912 German physicist Max von Laue discovered that when X-rays passed through a crystal sample diffraction and interferance occured. The effect was observed when a crystal was struck with a broad-band X-ray source and manifested as a ring-like pattern of dots on a photographic plate placed behind the crystal. As von Laue was investigating if X-rays were particles or waves, physicists William H. Bragg and William L.

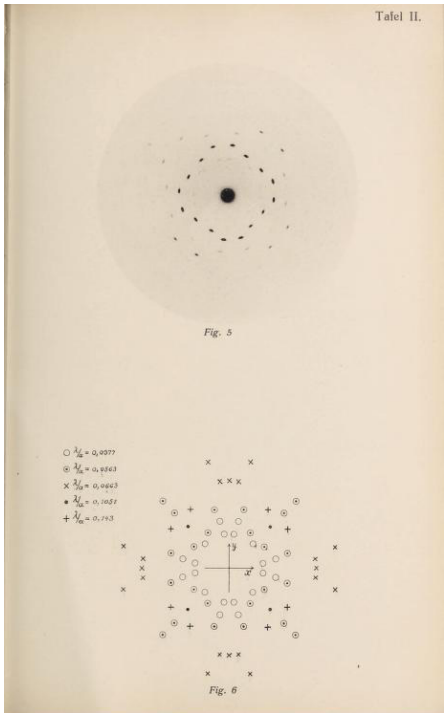


Figure 1: Interferance pattern from X-rays passing through a crystal sample [2]

Bragg investigated if the crystals could be used to reflect the X-rays.

Using an x-ray source, an adjustable apature and a turntable they were able to determine that scattering only occured when there was constructive interferance between the x-rays. They proposed that

$$n\lambda = 2d\sin\theta \tag{1}$$

was the condition for the reflected x-rays to be detected, where λ is the wavelength of the x-ray, d is the seperation between the atoms and θ is the angle of reflection.

More recently some of the assumption's that the original paper makes have been disputed as being ambiguous, particularily around the definition of the glancing angle. In 2025 physicist Pramode Bhattacharjee reformulated the relation to use the "unambiguous" angle of reflection instead:

$$n\lambda = -2d\cos\phi_1$$

where ϕ_1 is the unambiguous incidence angle. Because this is a compatively new discovery, the apparatus doesn't have markings for the unambiguous angle, so the older method was used.

2 Apparatus

- TEL-X-OMETER Type 2580 with
  - X-ray source
  - Crystal sample
  - Geiger-Meyer type radiation detector
  - Angle scale
- Philip Harris Digicounter

3 Procedure

The cable from the GM dectector was plugged into the "GM tube" port in the digicounter. The TEL-X-OMETER was closed and locked, before power was given to it. The tube voltage was set to 30 kV. The timer was set then the x-ray emmitter was enabled. Power was given to the digicounter.

For every 1° (from 20° to 115°), the count rate was measured using the digicounter. If the value seemed higher then it's neighbours the count rate was measured roughly every 10 arcminutes in either direction. Any fine adjustments were done using the thumb wheel.

4 Results

Seperation of NaCl atoms:

$$\lambda n = 2d \sin\theta$$
$$\lambda = 0.138 \times 10^{-9}\,\mathrm{m}, \quad n = 1, \theta = 0.584\,\mathrm{rad}$$
$$d = \frac{\lambda n}{2\sin\theta}$$
$$d_1 = 1.25 \times 10^{-10}\,\mathrm{m}$$
$$d_2 = 2.47 \times 10^{-10}\,\mathrm{m}$$
$$d_3 = 2.51 \times 10^{-10}\,\mathrm{m}$$
$$d_{avg} = 2.08 \times 10^{-10}\,\mathrm{m}$$

Lattice constant of NaCl:

$$(6.41 \pm 0.41) \times 10^{-10}\,\mathrm{m}$$

5 Error analysis

$$\Delta 2\theta = \frac{1}{6}^\circ$$
$$\Delta R = 0.1\mathrm{Count}/s$$
$$\Delta d = 2.08 \times 10^{-10}\,\mathrm{m}\sqrt{\left(\frac{\frac{1}{6}^\circ}{20^\circ}\right)^2 + \left(\frac{0.1\mathrm{Count}/s}{302\mathrm{Count}/s}\right)^2}$$
$$\Delta d = 1.73 \times 10^{-12}\,\mathrm{m}$$

6 Discussion

The count wheel only had markings part-way around its surface. There were approximate peaks where the markings did not proplery line up with the main scale or where the markings were fully missing. If we were repeating the experiment we would use a digital angle scale with a higher perscison. The lattice constant accepted value found to have a lattice constant of $a = 5.58 \times 10^{-10}$ m and a $d = 2.79 \times 10^{-10}$ m [4], compared to our measured values of $a = (6.41 \pm 0.41) \times 10^{-10}$ m and $d = (2.080 \pm 0.017) \times 10^{-10}$ m. The error in the seperation is likely caused by the missing units on the thumb dial.

7 Conclusion

By measuring the angle of refraction the values of the lattice constant and distance between atoms of $NaCl$ could be found.The value of d was found to be $d = (2.080 \pm 0.017) \times 10^{-10}$ m, which is outside the experimental error, likely due to some missing measurements on the thumb screw. The lattice constant was found to be $a = (6.41 \pm 0.41) \times 10^{-10}$ m, which is within the accepted value.

8 References

1. William Henry Bragg, William Lawrence Bragg, 1913, *The reflection of X-rays by crystals*, Proc. R. Soc. A1 July 1913

2. W. Friedrich, P. Knipping, M. Laue, presented by A. Sommerfeld 1912, *Interference Phenomena for X-Rays, Theoretical section*, Proceedings of the Mathematical-Physical Class of the Royal Bavarian Academy of Sciences in Munich, Issue 2. English translation by J. J. Stezowski.

3. Pramode Ranjan Bhattacharjee, *Discovery of the lack of success of Bragg's law in respect of perfect quantification of the X-ray diffraction phenomenon,* Next Research, Volume 2, Issue 2, 2025, 100312, ISSN 3050-4759

4. Chemical Book, *https://www.chemicalbook.com/article/sodium-chloride-crystal.htm*, 2023